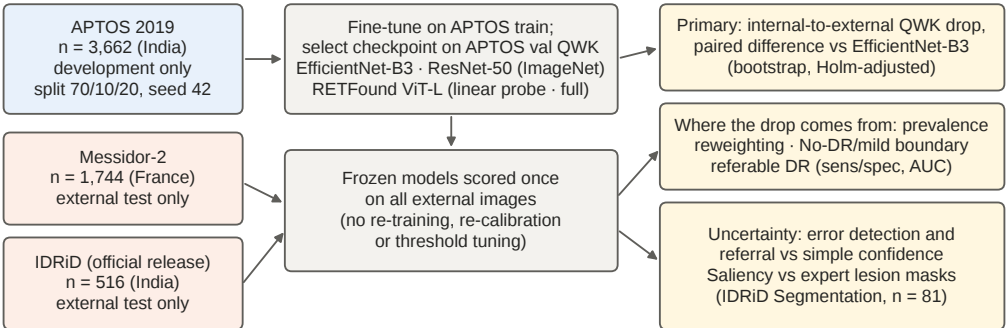
APTOS 2019
n = 3,662 (India)
development only
split 70/10/20, seed 42
Messidor-2
n = 1,744 (France)
external test only
IDRiD (official release)
n = 516 (India)
external test only
Fine-tune on APTOS train;
select checkpoint on APTOS val QWK
EfficientNet-B3 · ResNet-50 (ImageNet)
RETFound ViT-L (linear probe · full)
Frozen models scored once
on all external images
(no re-training, re-calibration
or threshold tuning)
Primary: internal-to-external QWK drop,
paired difference vs EfficientNet-B3
(bootstrap, Holm-adjusted)
Where the drop comes from: prevalence
reweighting · No-DR/mild boundary
referable DR (sens/spec, AUC)
Uncertainty: error detection and
referral vs simple confidence
Saliency vs expert lesion masks
(IDRiD Segmentation, n = 81)